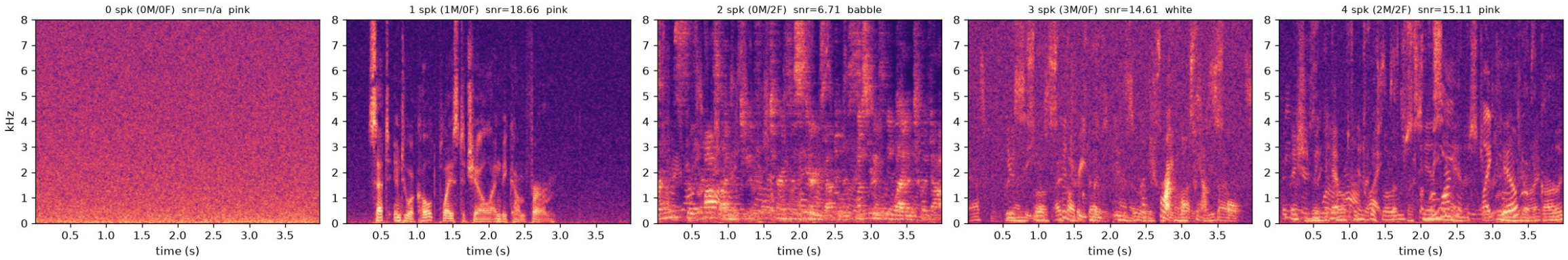
# Counting voices in a noisy room

A neural network that listens to 4 seconds of audio and reports how many people are talking, and how many are male or female




Five generated clips as the network sees them: 0, 1, 2, 3 and 4 people talking over background noise

**1** **Data**

a public speech corpus, turned into 20,000 synthetic clips

**2** **Input**

each clip becomes a mel spectrogram

**3** **Model**

a small CNN trained and tested on unseen voices

**4** **Experiment**

mel vs plain spectrogram as input

**5** **Real data**

tested and retrained on real meeting recordings

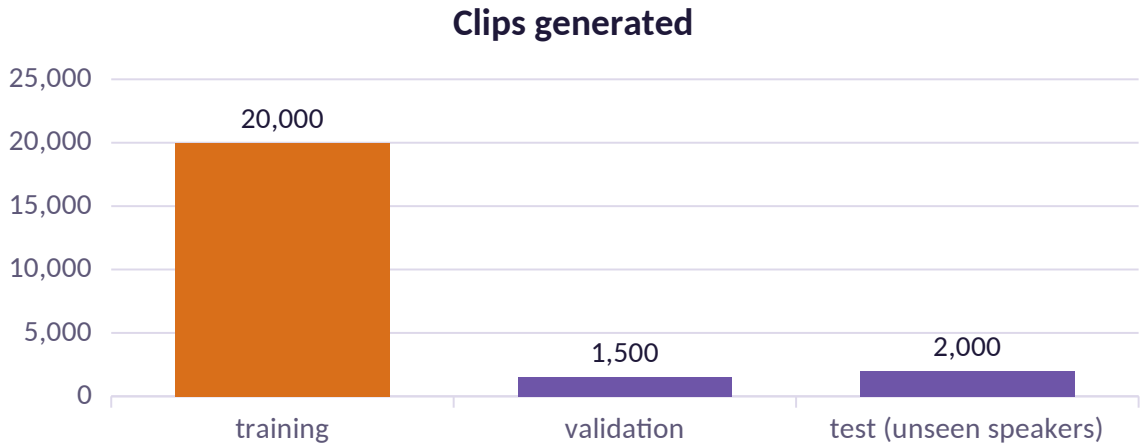
# Step 1: a public speech corpus, turned into 20,000 synthetic noisy clips

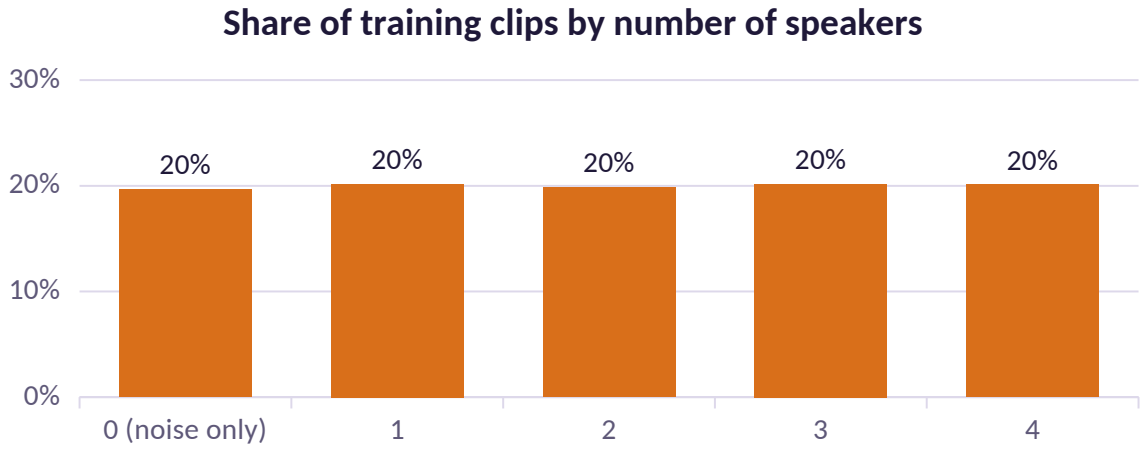
## Source: LibriSpeech

- Free public corpus of audiobook readings (LibriVox), widely used in speech research
- 291 speakers used, 146 male / 145 female, each recorded alone in a quiet room
- No overlapping speech and no noise, so it cannot be used directly

## Generation recipe

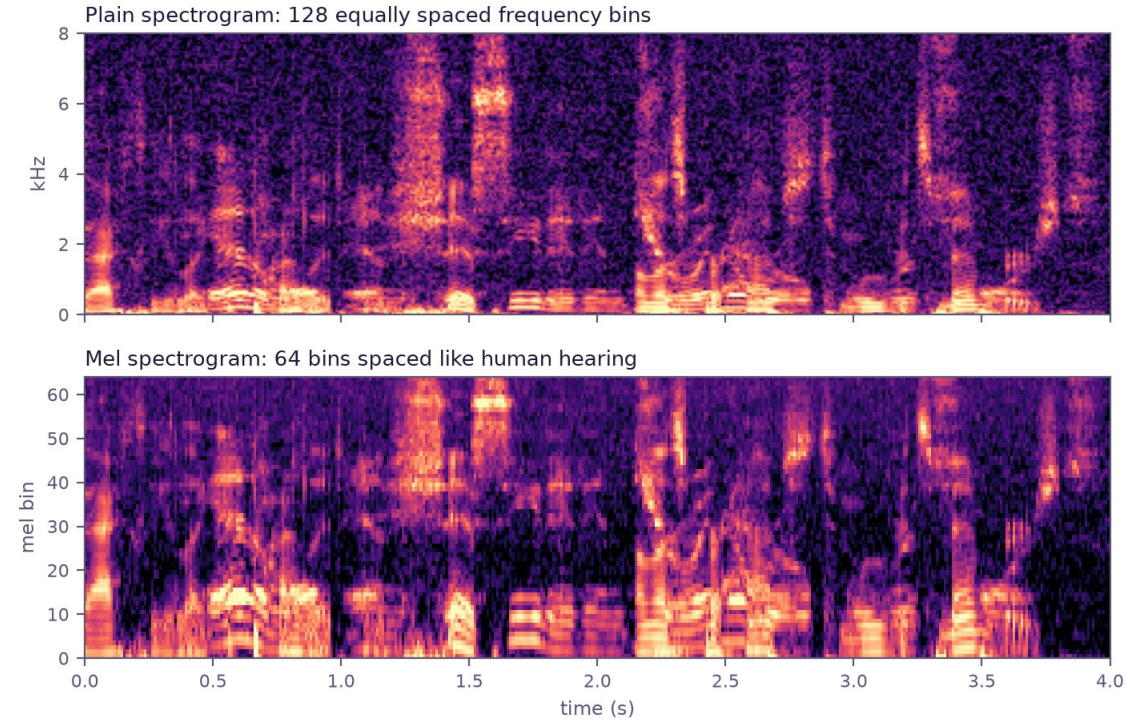
- Pick 0 to 4 speakers; overlay a 4-second excerpt from each
- Add reverb with probability 0.5; add white, pink or babble noise at 0 to 20 dB SNR
- Labels (people, male, female) are exact by construction; test speakers are never seen in training



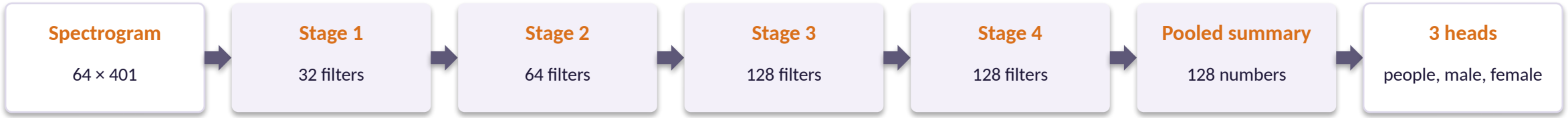



# Step 2: each clip becomes a spectrogram, read by a small convolutional network

- **Spectrogram:** time left to right, frequency bottom to top, brightness is energy. A voice is a stack of horizontal harmonics; two voices interleave.
- **Mel scale:** 64 bands spaced like human hearing, fine at low frequencies and coarse at high. The plain version (top) keeps 128 equal bins; I compare both in step 4.
- **Model:** a CNN, the standard architecture for images. Four stages detect patterns and halve the picture, then three output heads each pick a count from 0 to 4. About 580,000 parameters.




Same clip: 2 speakers, white noise at 20 dB SNR, no reverb

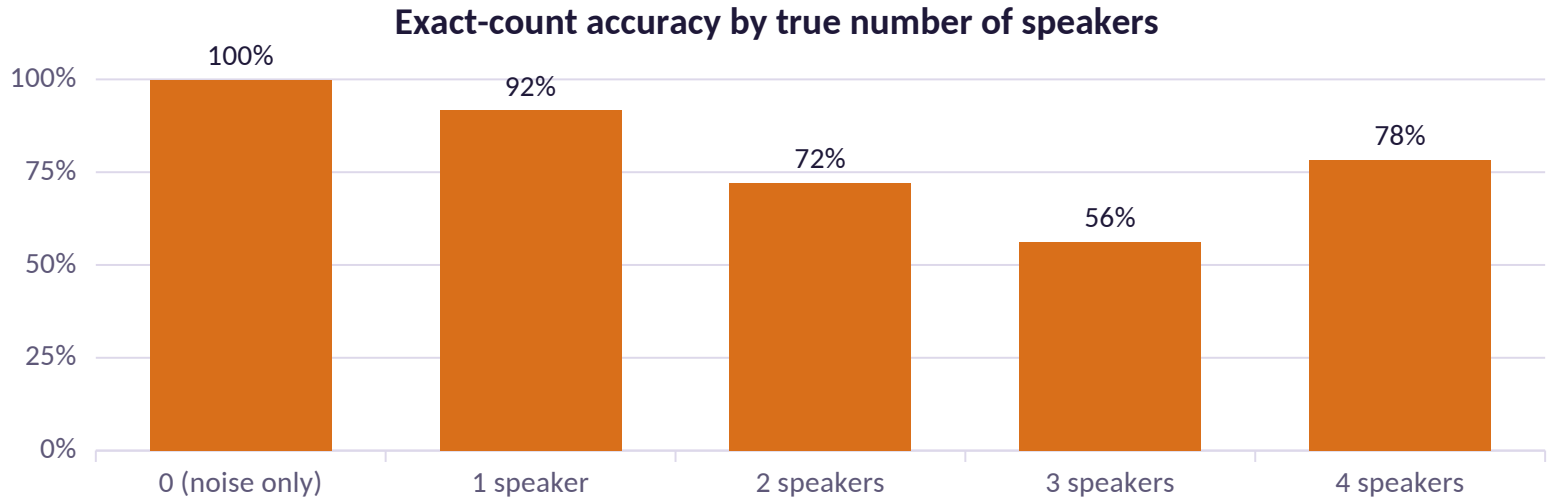


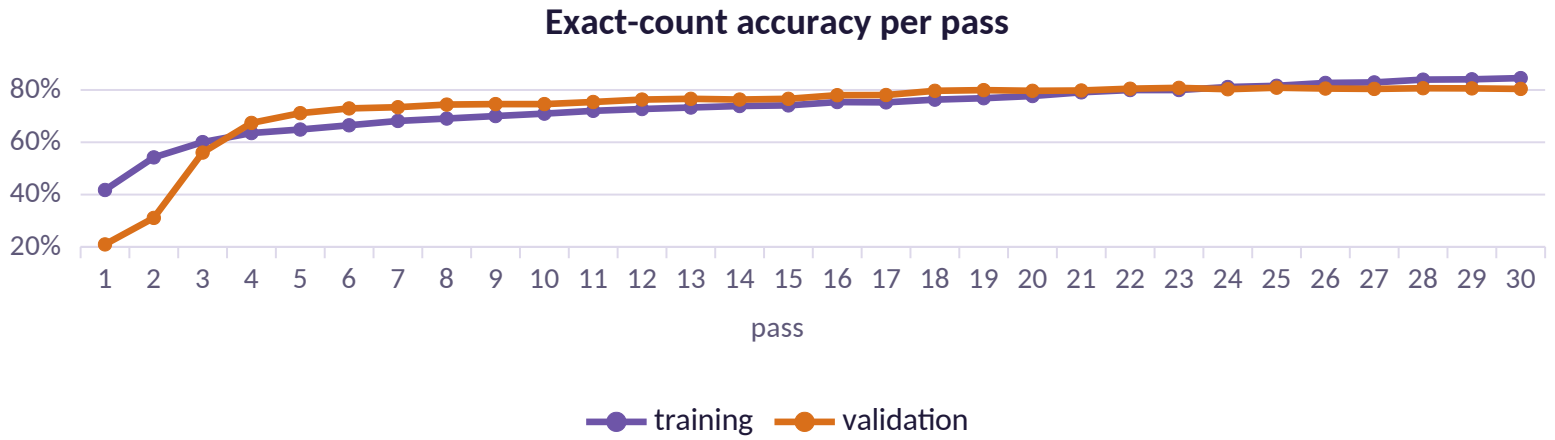
# Step 3: trained for 30 passes, tested on 2,000 clips from unseen speakers

**80%**
exact speaker count

**99%**
within one speaker

**75% / 78%**
exact male / female count

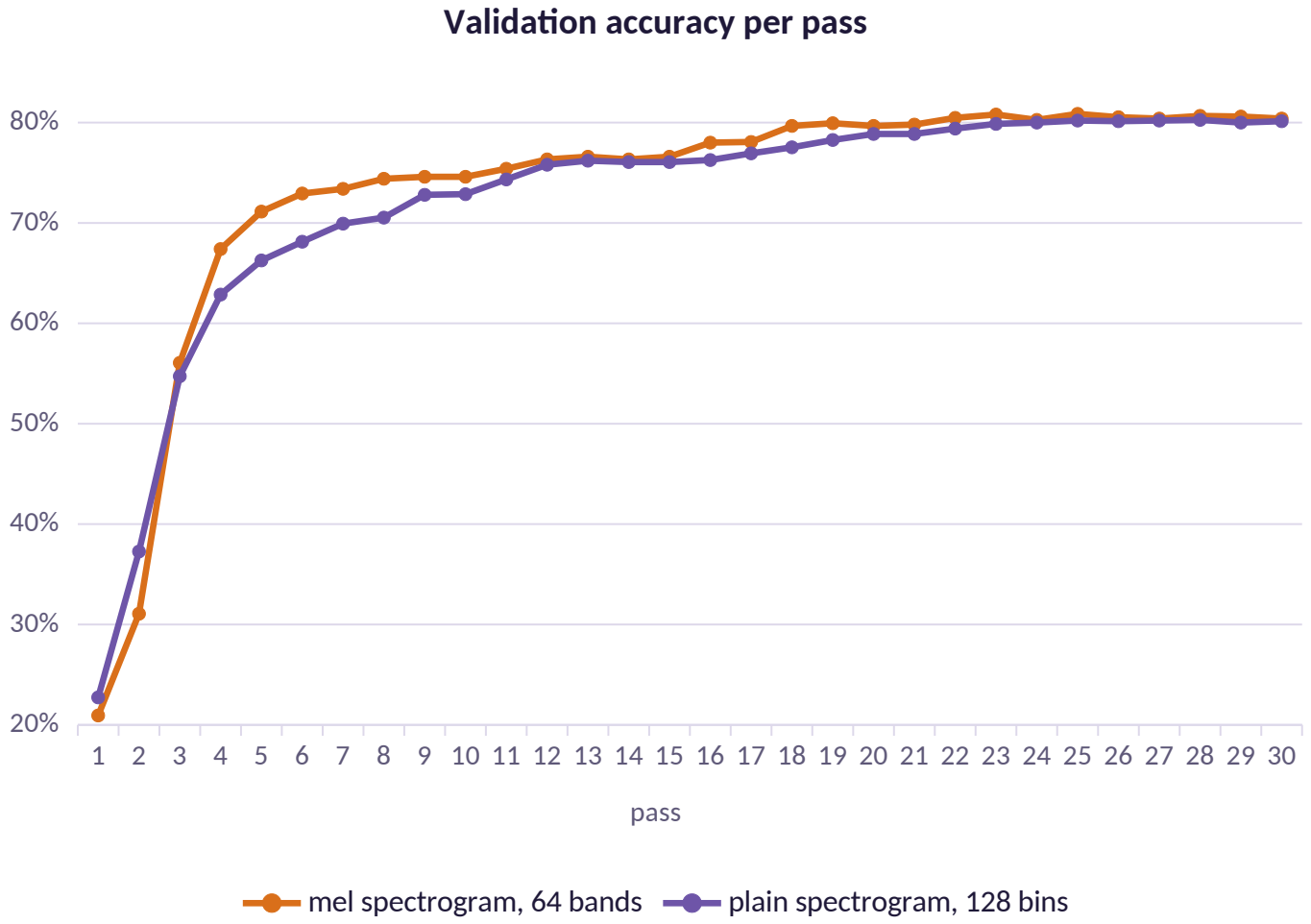



- 30 passes over the training clips; the version with the best validation score is kept
- About 1.5 h per model on a laptop GPU

Hardest case: three simultaneous speakers (56%). Low SNR, reverb and babble noise hurt most.

# Step 4: mel vs plain spectrogram as input, everything else identical




| | mel | plain |
|---|---|---|
| input size | 64 × 401 | 128 × 401 |
| exact count, 2,000-clip test | **80.0%** | **79.2%** |
| exact count, 1,000-clip test | **83.4%** | **81.4%** |
| within one speaker | 98.6% | 98.5% |
| time per pass | 165 s | ~320 s |

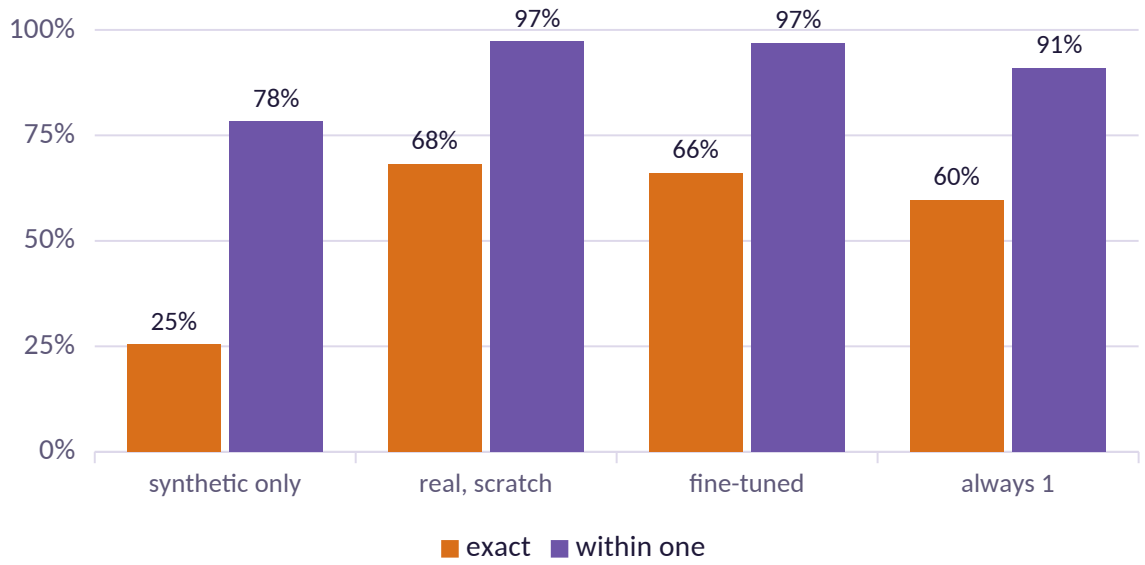
- **Only the input changed:** same clips, network, settings and random seed.
- **Mel is 1 to 2 points better** and trains twice as fast, since its input is half the size.
- **One run each,** so the gap is suggestive, not proven.

# Step 5: on real meeting recordings the synthetic model collapses; retraining on real data recovers it

- **Real data:** the AMI Meeting Corpus, about 100 h of real 4-person meetings with time-stamped speaker turns and participant sex. Each 4 s window gets exact labels from the turns.
- **Setup:** 42,000 balanced training windows from 136 meetings (far-field table mic and headset mix); 8,000 test windows from 16 unseen meetings at the natural distribution.
- **Same network, same recipe;** only the training audio changed.

| model | trained on | exact | within one |
|---|---|---|---|
| **cnn_v4** | synthetic clips only | **25%** | **78%** |
| **cnn_v6** | real meetings, from scratch | **68%** | **97%** |
| **cnn_v7** | cnn_v4 fine-tuned on real | **66%** | **97%** |
| **baseline** | always answer "1" | **60%** | **91%** |

**Speaker-count accuracy on 8,000 real meeting windows**



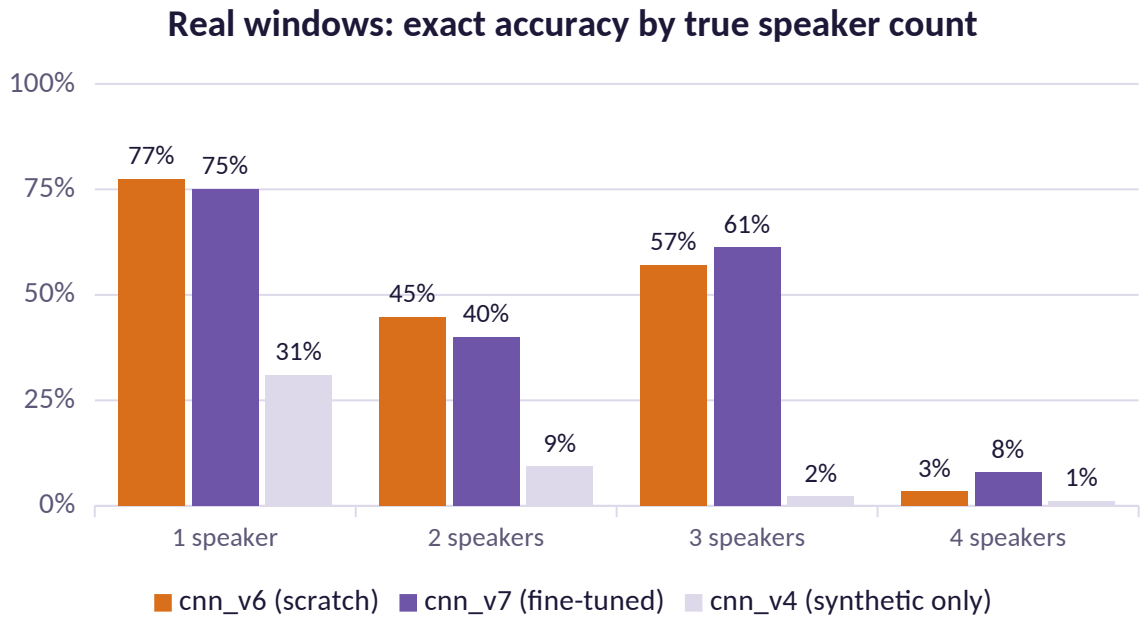

- **Why cnn_v4 fails:** it labels most real speech as silence. Far-field room audio is quieter and more echoey than the synthetic clips.
- **Retrained model by true count:** 0: 98%, 1: 77%, 2: 45%, 3: 57%, 4: 3% (only 89 four-speaker windows exist).
- **Fine-tuning from cnn_v4 did not help** and that model fell to 38% on the synthetic test: it forgot the old domain.

# Step 5, by count: both real-data models fail on 2 and 4 speakers

- **Same 8,000 real test windows,** split by the true number of speakers. Natural distribution, so one speaker dominates.
- **cnn_v6, from scratch:** random starting weights, 20 passes over the 42,500 real windows, learning rate 5e-4.
- **cnn_v7, fine-tuned:** starts from cnn_v4's trained weights, then 8 passes over the same real windows at a lower learning rate (2e-4), so the synthetic knowledge is adjusted rather than overwritten. All layers updated, no synthetic clips mixed in.

| model | 1 speaker | 2 speakers | 3 speakers | 4 speakers |
|---|---|---|---|---|
| windows | 4,768 | 1,975 | 642 | 89 |
| cnn_v6 (scratch) | 77% | 45% | 57% | 3% |
| cnn_v7 (fine-tuned) | 75% | 40% | 61% | 8% |
| cnn_v4 (synthetic) | 31% | 9% | 2% | 1% |

Exact count accuracy on the natural AMI test set. Orange marks the better of the two real-data models at each count. Silent windows (526) are near 98% for both and are left out.




- **One speaker:** the scratch model wins, 77% vs 75%, and this is most of a real meeting.
- **Two speakers is the real weakness:** 45% and 40%; most misses are heard as one person, since real overlaps are brief.
- **Four speakers:** only 89 windows, and 80 of them are counted as three. Off by one, not lost.

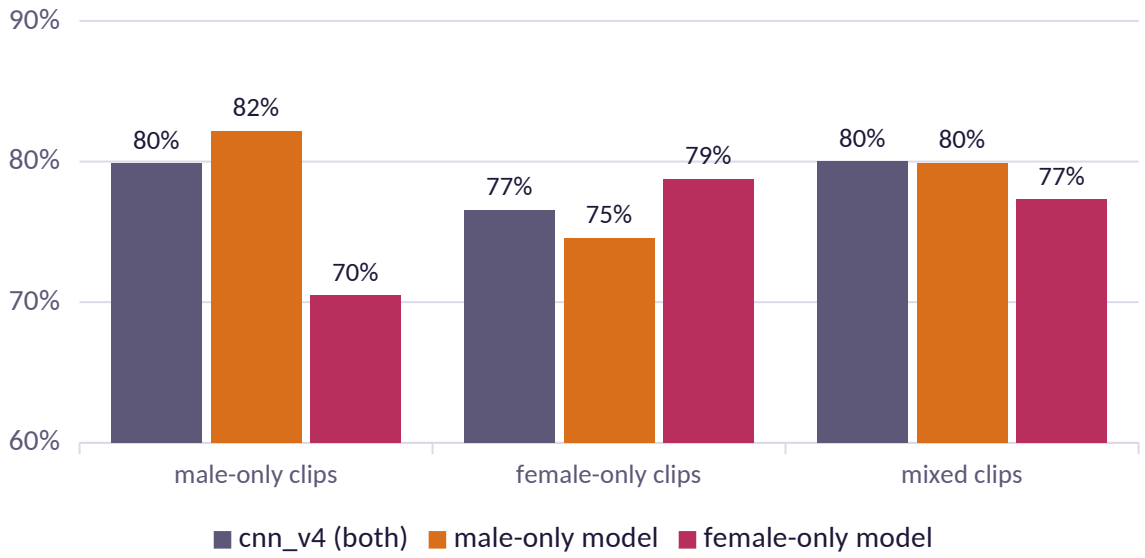
# Step 6: single-sex training helps a little, and only for that sex

- **Question:** is counting easier when every voice in the clip is the same sex?
- **Setup:** two new synthetic sets with the same recipe, one from the 146 male speakers only and one from the 145 female speakers only; 20,000 training clips each, separate test speakers.
- **Same network, same recipe;** three models compared on three test sets.

| model | trained on | male test | female test | mixed test |
|---|---|---|---|---|
| **cnn_v4** | both sexes | **80%** | **77%** | **80%** |
| **cnn_v8** | male voices only | **82%** | **75%** | **80%** |
| **cnn_v9** | female voices only | **70%** | **79%** | **77%** |

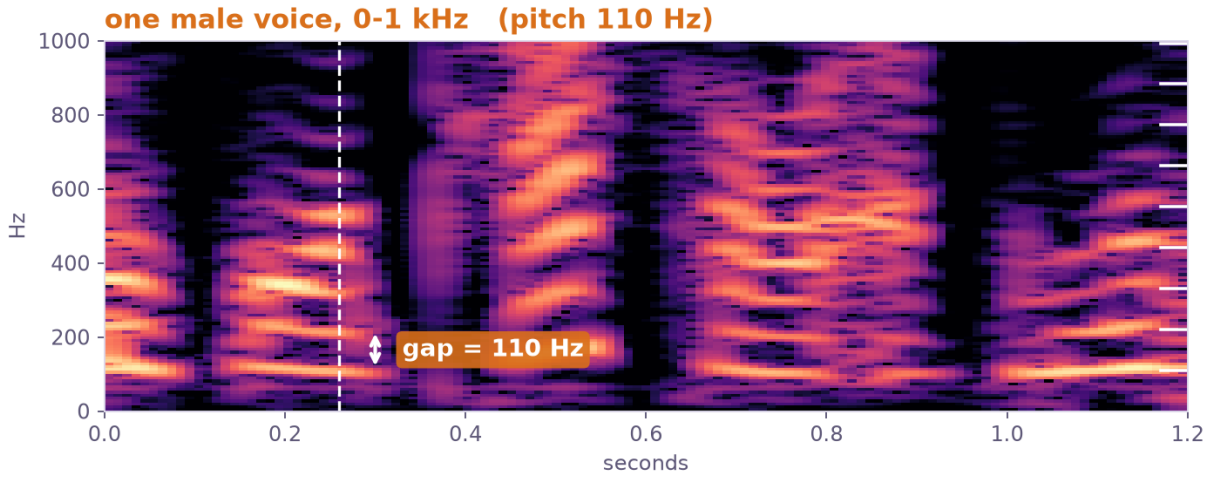
Exact speaker-count accuracy, 2,000 clips per test set. Within-one accuracy is 96 to 98% for every pair.

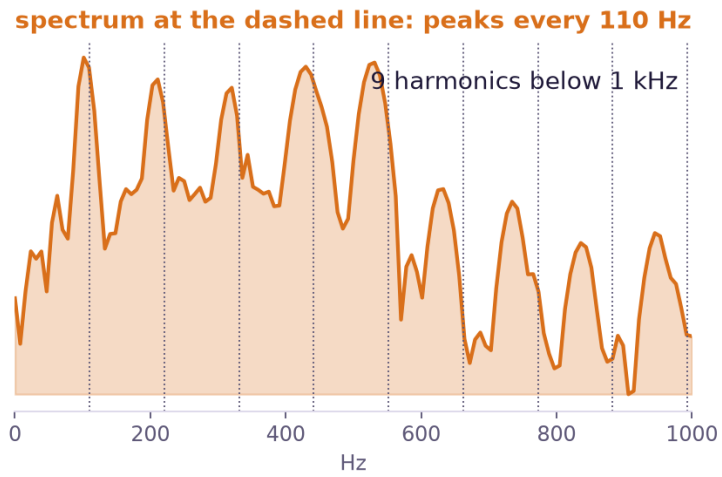
**Exact count accuracy by test set**

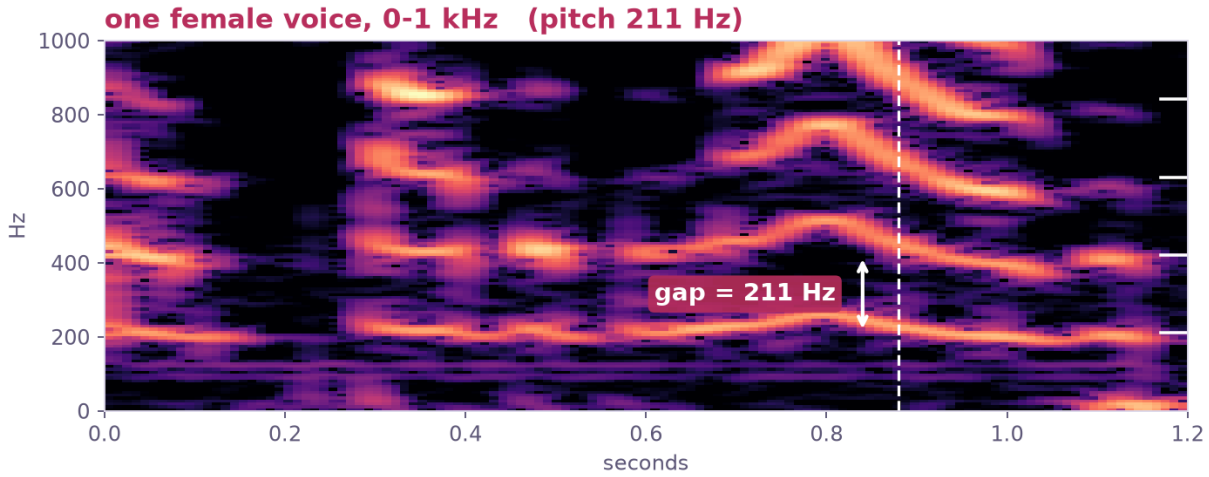


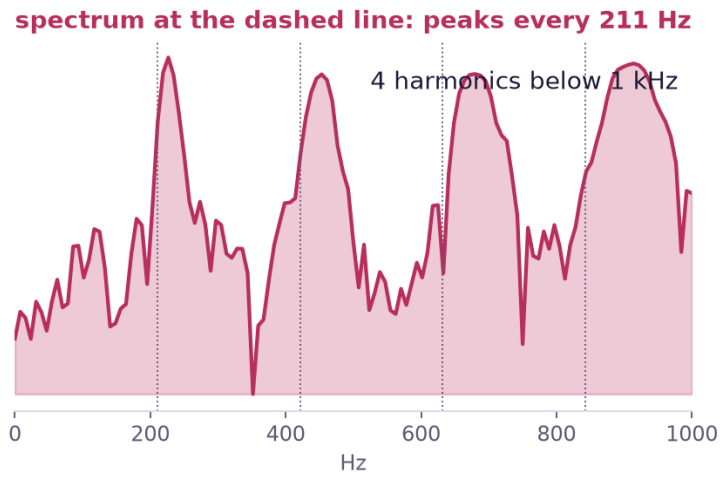

- **Specialists win on their own sex,** by 2.3 points (male) and 2.2 points (female): a real but small gain.
- **They do not transfer:** each loses 5 to 9 points on the other sex, so one mixed model is the safer choice.
- **Female voices are harder for every model:** two or three overlapping female voices score about 10 points below the same counts for male voices.

# Step 6, continued: male vs female voices on the spectrogram



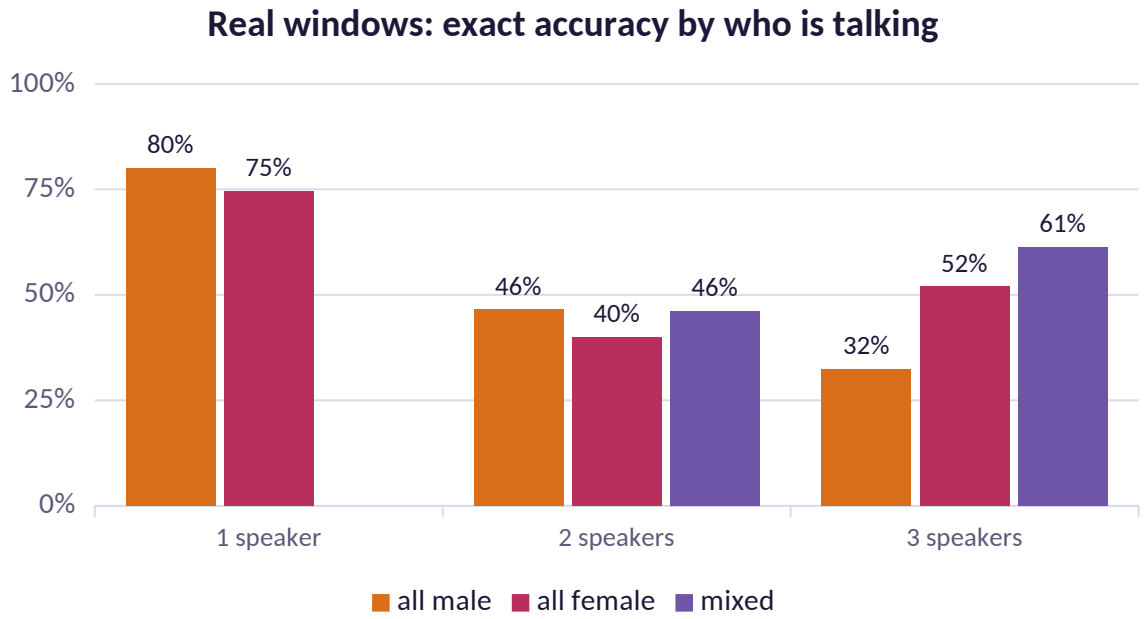

A voice is a stack of harmonics spaced by its pitch. Male: ~110 Hz apart, 9 lines below 1 kHz. Female: ~210 Hz apart, only 4. Clean LibriSpeech audio, 100 ms analysis window.

# Step 6 on real data: how many talk matters more than who

- **Same real-data model (cnn_v6),** 8,000 real meeting windows split by who was talking: only men, only women, or both. No retraining.
- **Mixed windows always contain 2 or more speakers,** so compare at a fixed count, not overall.

| windows | overall | 1 speaker | 2 speakers | 3 speakers |
|---|---|---|---|---|
| **all male (3,202)** | 72% | **80%** | **46%** | **32%** |
| **all female (2,832)** | 68% | **75%** | **40%** | **52%** |
| **mixed (1,440)** | 49% | **n/a** | **46%** | **61%** |

Exact count accuracy of cnn_v6 on the natural AMI test set. The 3-speaker single-sex groups have only 71 and 77 windows; 4-speaker windows are too rare to score.




- **Female-only is hardest on real data too:** 75% vs 80% for one speaker, 40% vs 46% for two.
- **Mixing sexes does not hurt:** at 2 speakers mixed equals all-male; at 3 it is the best group. A male and a female voice sit in different pitch ranges and are easier to tell apart.
- **Overall gaps are mostly about count:** real meetings are mostly one person at a time, and single-speaker windows are the easy ones.